James Ridey

Software Engineer | Frontend, Product Engineering & Performance

Sydney, Australia • Australian citizen (E-3 visa eligible) • Seeking relocation to the San Francisco Bay Area
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Software engineer with frontend experience at Nullify, Canva, and Google. Looking for frontend or broader software work with difficult problems to solve and room to learn.

Technical Skills

Professional: TypeScript, JavaScript, React, Go, Python, AWS.

Frontend & tooling: Vite, Playwright, Storybook, ESLint, RAG interfaces, LLM-assisted development.

Projects: C/C++, CUDA, Verilog, Arduino, ESP32, STM32, PCB design.

Professional Experience

Nullify Product Engineer | AI application security — Mar 2026 – present

- Owned UI2's frontend rebuild and built most features, using Vite and atomic design to organise about **470 components**.
- Set the design direction and used Claude Design to try out ideas.
  Made the UI faster and more consistent, improved mobile support and navigation, and raised accessibility scores.
- Built Playwright/Storybook screenshot comparisons with ImageMagick. Added end-to-end tests for UI flows.
  Used an LLM to build ESLint rules for dependencies, colour tokens, and shared UI elements.
- Investigated slow backend startup and improved performance in Go and Python services.

Canva Software Engineer — 2020 – 2025

- Searched Canva's codebase for a way to render Magic Design previews and adapted the editor's canvas rendering.
  Agreed on reuse with the editor team and took responsibility for the resulting technical debt.
- Built Team Brain's frontend with a backend engineer: a working RAG prototype answering questions from documents.

Google Software Engineer Intern — 2017 – 2019; two summer internships

- Reverse-engineered a factory-test app and found command injection in its network diagnostic, enabling development access on a retail KaiOS phone. Ported Neighbourly under tight app-size and 512 MB RAM constraints; wrote the team's first porting specification.

ThoughtDesign Junior Software Engineer — 2014 – 2017

Independent Projects & Contributions

OnnxStream & E-paper Art Frame C++ inference & Raspberry Pi

- Added SDXL Turbo support to OnnxStream, adapting tensor dimensions and tiled decoding; fixed tokenisation and VAE scaling bugs. **Maintainers accepted five PRs**.
- Built a Raspberry Pi e-paper frame that generates images locally every day, with a Python service, web controls, and colour processing for the display.

Minecraft pack.png Seed Search CUDA & GPU profiling

- Used NVIDIA Visual Profiler to find early thread exits in Minecraft's pack.png seed search. Splitting filtering and tree-position work into separate CUDA kernels made the search run **roughly twice as fast** on my machine.

JRB8 Computer Digital logic, Verilog & toolchain

- Designed an 8-bit CPU fabricated through Tiny Tapeout; built its instruction set, assembler, language, compiler, and browser-based compile-flash-run tool.

Brother Embroidery Cartridge Hardware & software reverse engineering

- Built a working replacement cartridge through reverse engineering; published findings and code reused by others.

Security Credentials

Flare-On: Completed all challenges in 2022–2025; **157th / 4,100+** entrants in 2025.

Burp Suite Certified Practitioner (2026) • DEF CON Offensive IoT Exploitation course.

Education

Macquarie University BEng (Honours), Software Design — 2016 – 2020